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Textile Microplastics – a brief update

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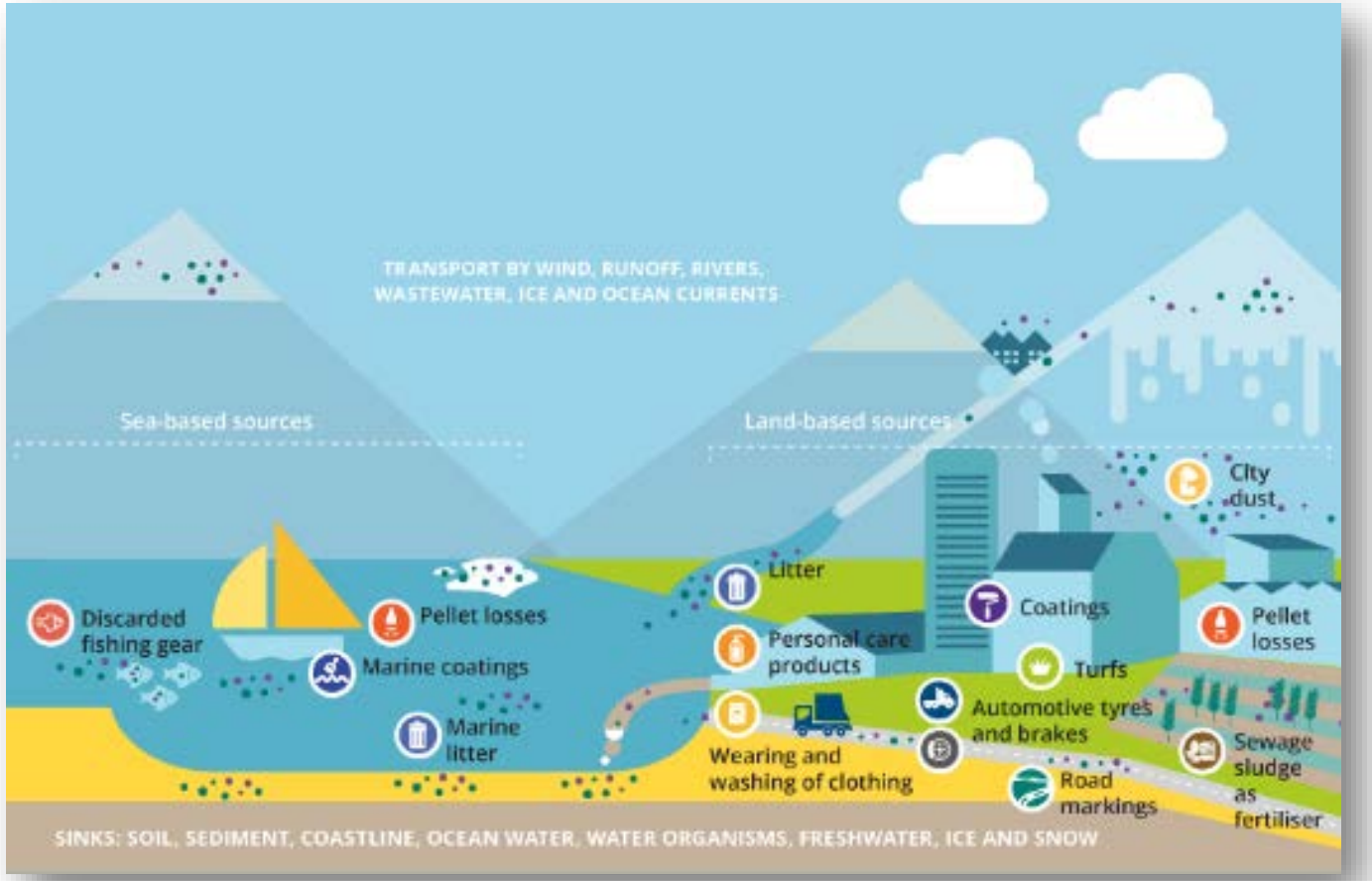
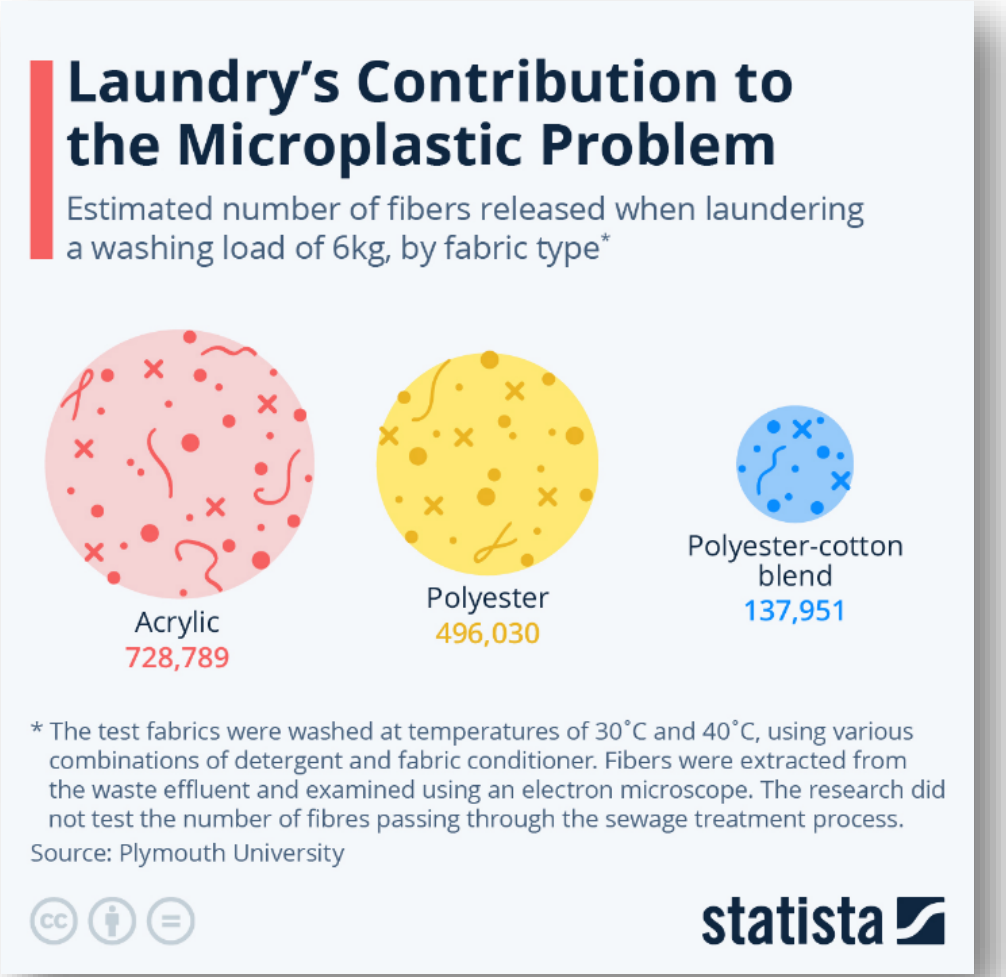
WOLF IN NYLON CLOTHING

- Synthetic clothing a huge contributor to microfibre pollution
- Serious concern for India

- Awareness in India regarding concerns of synthetic textiles negligible
- Textile industry contributes 2% to India's GDP

Textile microplastics





Microplastics shedding during use

Clothing and home textiles

- Household laundry
- Heterogeneous number of washing cycles
- Very individual and uncontrolled

Commercial use / rental

- 25-100+ cycles commercial laundry
- Controlled processes
- Mainly PES (and CO) used

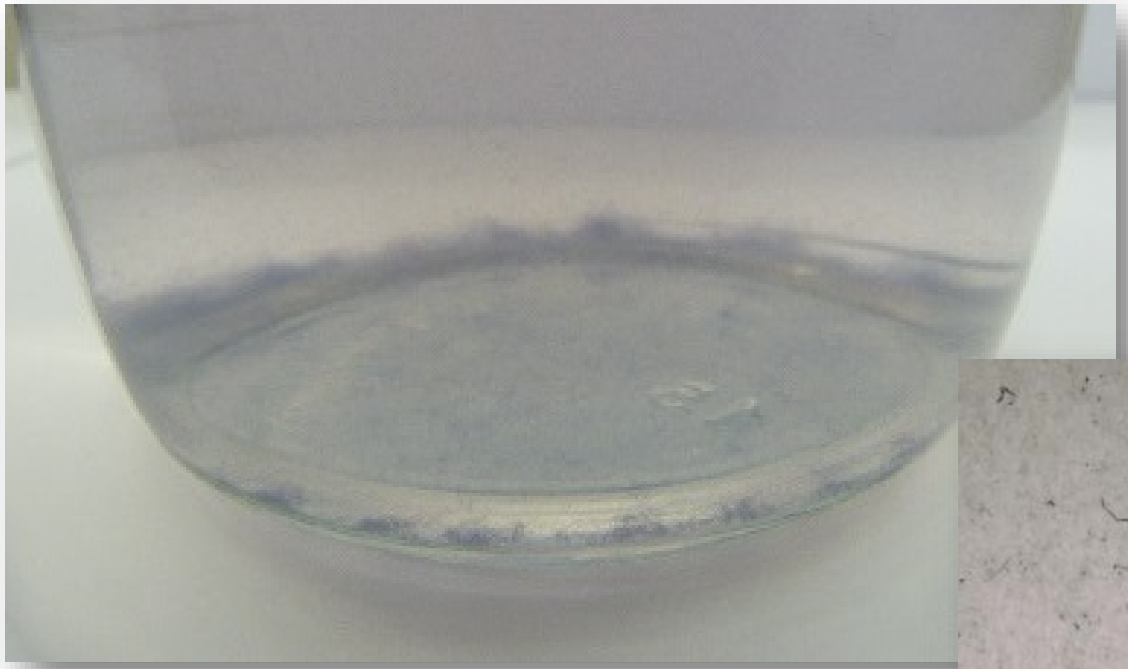
Technical textiles

- Very limited data so far
- Normally not re-processed
- Some products remain in the environment

Microfibers, microparticles, microplastics?

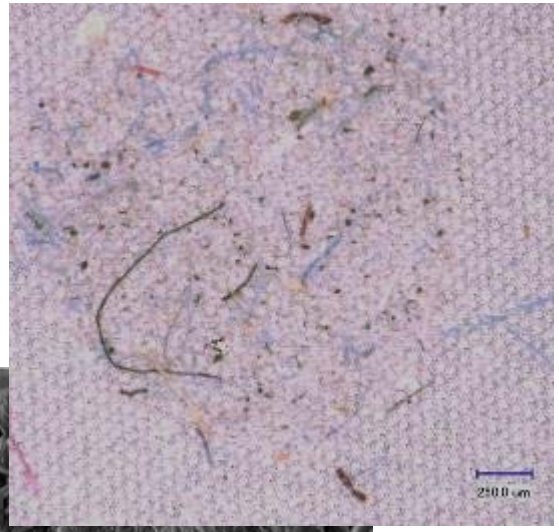
What is really microplastics?

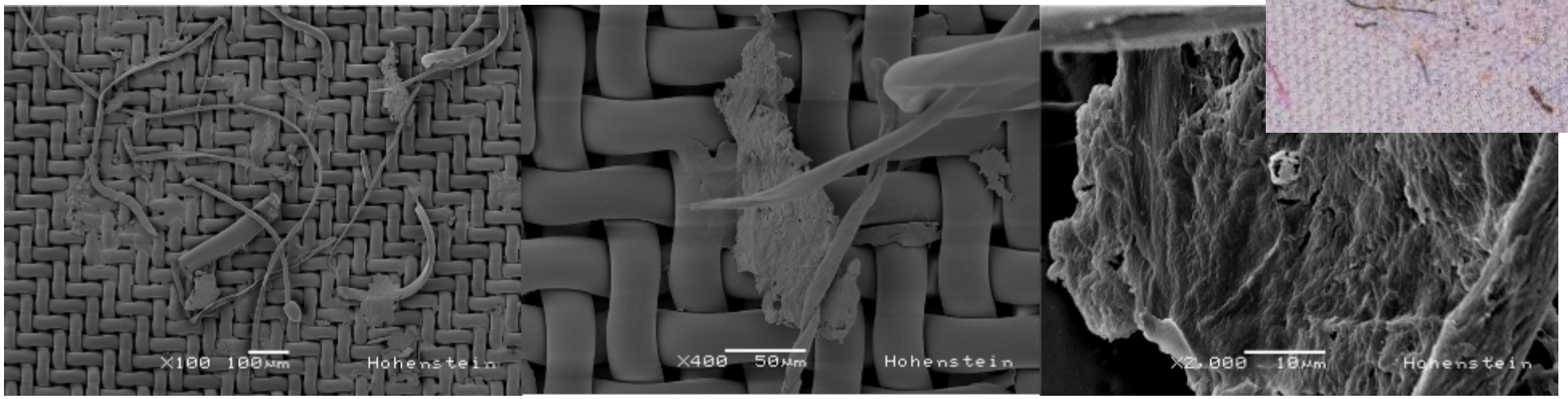
Do we all know, what we are talking about?



Closer look on a wastewater sample

- Fibers with different colours
- Different sizes and diameters
- Polyester and cotton fibers
- Contaminations





Fiber analysis

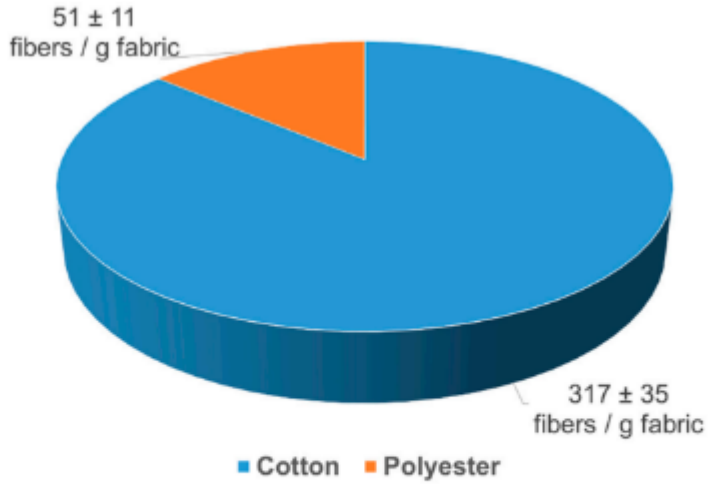


Figure 7. Fiber counts per g fabric of polyester and cotton fibers regarding the total amount of shedded fibers. These results refer to a wastewater sample from a washing process of a 50/50 polyester-cotton intimate blended fabric analyzed by dynamic image analysis.

water MDPI

Article

Microplastic Fibers Released by Textile Laundry: A New Analytical Approach for the Determination of Fibers in Effluents

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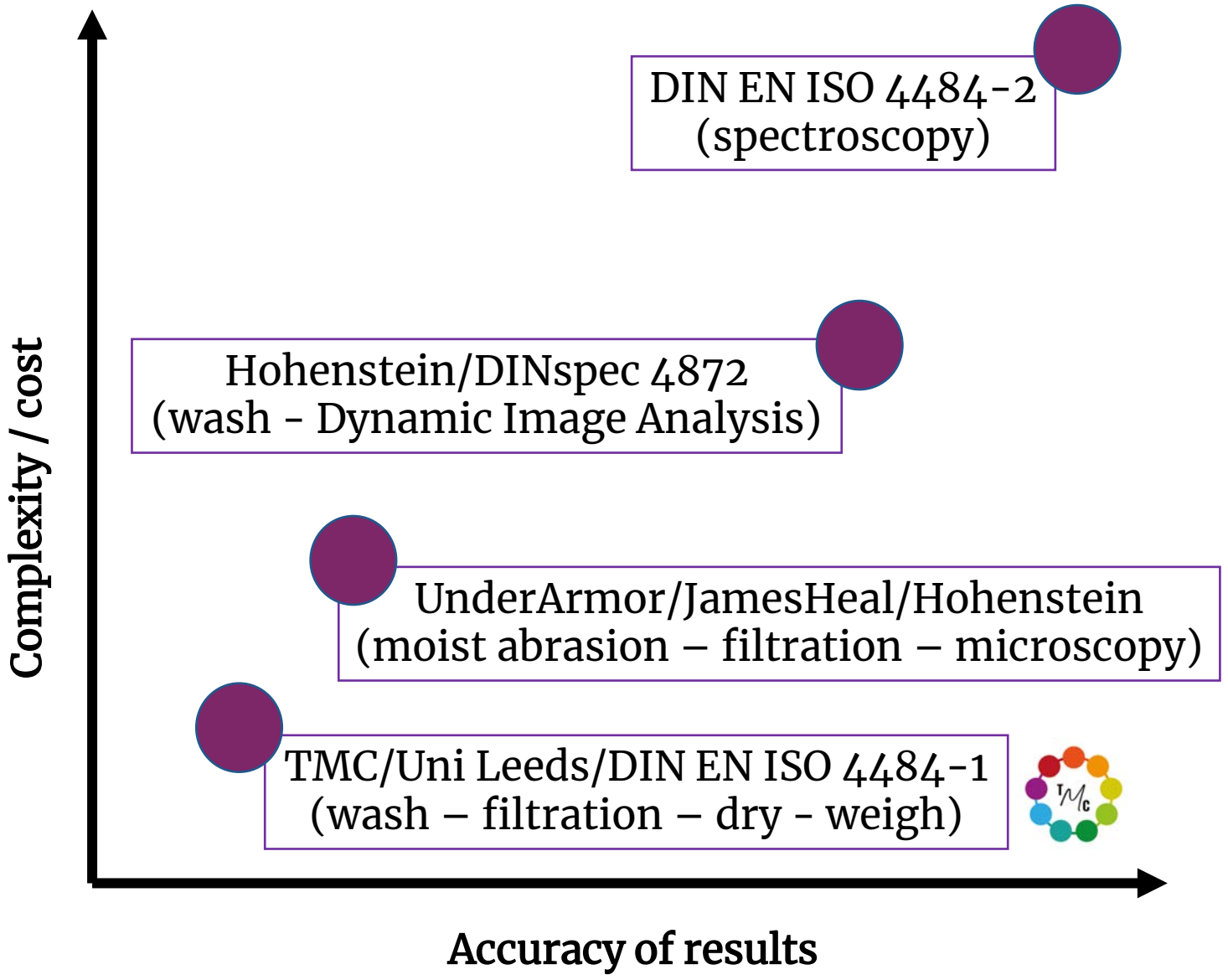
Test methods for microplastic release from textiles



Limitations

- Public discussion is still emotion-driven, not fact-driven
- Only little benchmarking data available
- Commercial use / home laundry different
- Which test method will be the leading one?
- No thresholds/limit values so far

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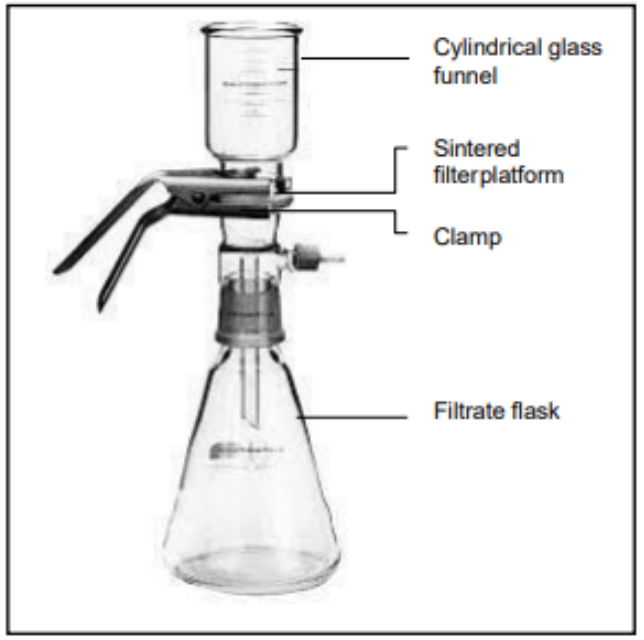
TMC - gravimetric analysis

- The Microfiber Consortium (TMC)
- Sample preparation and standardized washing process (Gyrowash containers) without detergent
- Filtration and weighing process follow

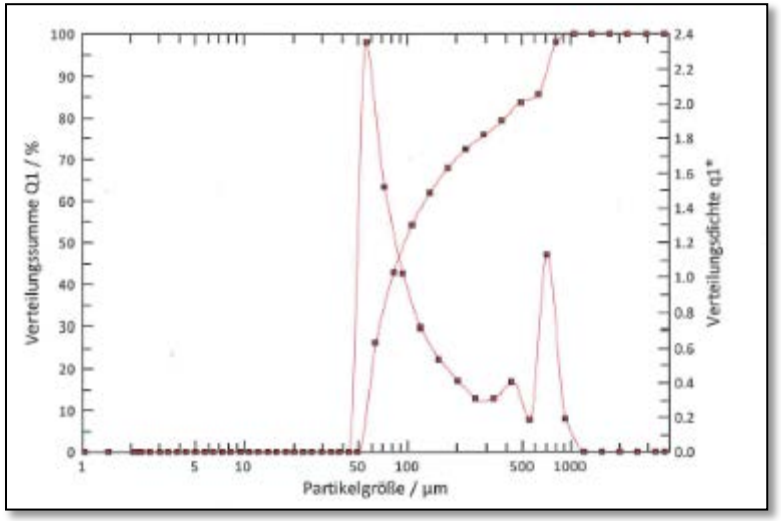
- The fibre release and entire residue (consisting fibres and partickles) is assessed by comparison of the filter mass before and after testing.

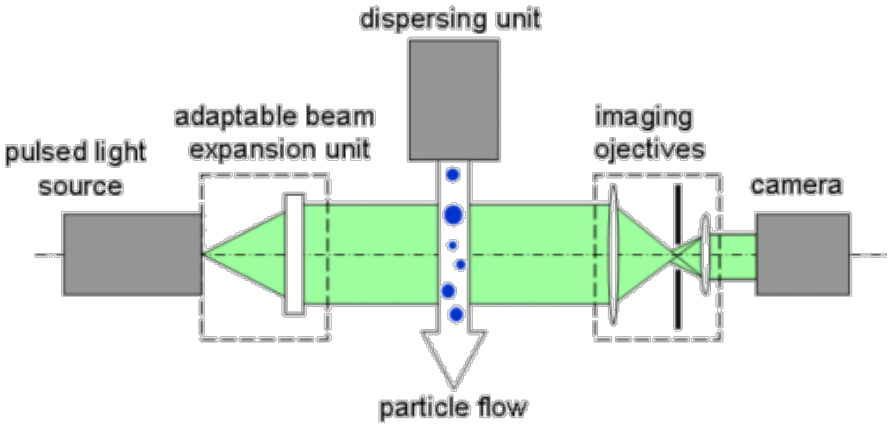


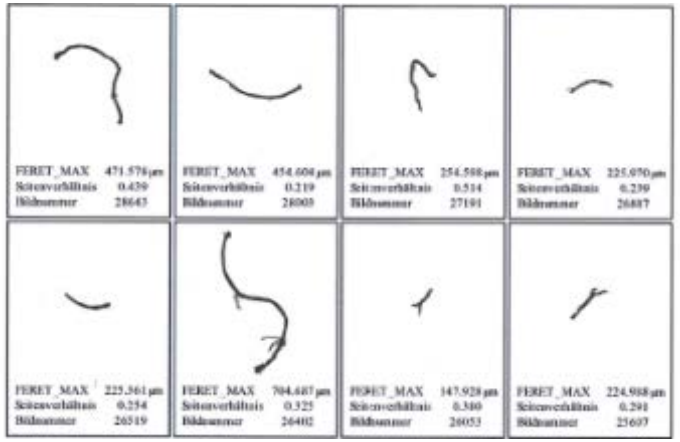
Dynamic Image Analysis (DIA)

particle size and shape, particle amount,
even in turbid and coloured liquids

DIA is a smart and highly precise method.







~ 10 000 images per measurement